| | | | |
|---|---|---|---|
| **Fiscal Year:** | FY 2006 | **Task Last Updated:** | FY 09/21/2005 |
| **PI Name:** | Masson, Patrick H. Ph.D. | | |
| **Project Title:** | Combined Proteomic And Molecular Genetic Approaches To The Study Of Early Phases Of Root Gravitropism In Arabidopsis thaliana | | |
| **Division Name:** | Space Biology | | |
| **Program/Discipline:** | SPACE BIOLOGY | | |
| **Program/Discipline--Element/Subdiscipline:** | SPACE BIOLOGY--Cellular and molecular biology | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | (1) Cell & Molecular Biology<br>(2) Plant Biology | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
| **PI Email:** | phmasson@wisc.edu | **Fax:** | FY 608-262-2976 |
| **PI Organization Type:** | UNIVERSITY | **Phone:** | 608-265-2312 |
| **Organization Name:** | University of Wisconsin Madison | | |
| **PI Address 1:** | Laboratory of Genetics (Room 3262) | | |
| **PI Address 2:** | 425G Henry Mall | | |
| **PI Web Page:** | https:// | | |
| **City:** | Madison | **State:** | WI |
| **Zip Code:** | 53706-1580 | **Congressional District:** | 2 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 2001 Space Biology 01-OBPR-06 |
| **Start Date:** | 12/01/2002 | **End Date:** | 11/30/2005 |
| **No. of Post Docs:** | 1 | **No. of PhD Degrees:** | 0 |
| **No. of PhD Candidates:** | 0 | **No. of Master' Degrees:** | 0 |
| **No. of Master's Candidates:** | 1 | **No. of Bachelor's Degrees:** | 0 |
| **No. of Bachelor's Candidates:** | 2 | **Monitoring Center:** | NASA ARC |
| **Contact Monitor:** | | **Contact Phone:** | |
| **Contact Email:** | | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | | | |
| **Grant/Contract No.:** | NAG2-1602 | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |
| | Gravitropism is an important process for plant growth and development. It dictates upward shoot growth and downward root growth, positioning both organs in environments that are adequate for their primary functions. Hence, long-term space-exploration missions, or colonization of other planets, will probably require that plants be engineered to better adapt to micro- and/or hypergravity. Unfortunately, this cannot be accomplished as long as we do not understand the processes that govern gravity signal transduction. Genetic and physiological strategies have identified several molecules involved in gravitropism. However, complete sequencing of the Arabidopsis thaliana genome revealed that most genes are members of gene families, suggesting that widespread functional redundancy might complicate forward genetics. There is a need for alternative approaches to the identification of new gravity signal transducers. Using two-dimensional gel electrophoresis, we identified root-tip proteins whose abundance, modification and/or cellular localization are | | |

| | |
|---|---|
| **Task Description:** | affected early after gravistimulation. Several of them were affected within the first 12 minutes of stimulation, preceding the curvature response. Among these, an unusual response regulator (ARR22) was up-regulated during this period. Typically, response regulators are implicated in signal transduction pathways that involve two-component receptors. This proposal will describe experiments aimed at identifying more of these proteins by mass spectrometry, and investigating their distribution in root-tip cells before and after gravistimulation. Mutations will be isolated in the corresponding genes (including ARR22), and their phenotypes will be analyzed. If/when a gravitropic defect is found, we will collaborate with other investigators to analyze the effect of these mutations on amyloplast sedimentation, the visco-elastic properties of the statocytes, the gravity-induced relocalization of auxin efflux carriers in the root tip, and cytoplasmic alkalinization of the statocytes early after gravistimulation. This multidisciplinary approach will fill some of the gaps that hinder our understanding of gravity perception and signal transduction in plants. Objective of this task is to identify and characterize root-tip proteins that are differentially expressed, modified or targeted early in response to gravity stimulation. While Genetics has been fruitful at identifying genes that contribute to gravity signal transduction in the root tip of plants, it also suffers from limitations due to functional redundancy. Proteomic strategies allow direct access to proteins whose abundance, localization and/or modification vary in response to gravity stimulation, regardless of gene redundancy. When such proteins are found, reverse genetics can then be used to characterize their involvement in gravity signal transduction. This proteomic approach complements microarray analyses of gene-expression responses to gravity stimulation because they can account for translational and post-translational modifications, which constitute potential regulatory mechanisms in gravity signal transduction. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | This project will allow us to better understand the molecular mechanisms that govern gravity signal transduction in plants. In turn, it may provide the knowledge and tools that are needed to engineer plants that display a better ability to respond to lodging by wind and rain, a major source of crop loss in the world. Furthermore, the new proteomic approaches that are being developed as a part of this project are likely to be very useful in the study of multiple developmental processes and responses to environmental stimuli in plants. |
| **Task Progress:** | Gravity is a ubiquitous directional cue used by most plant organs, including roots, to guide their growth. As such, it conditions plant growth, development and adaptation to various environments. Yet, little is known about the molecular mechanisms that govern gravity perception and signal transduction in plants. We have used a proteomic approach in Arabidopsis thaliana to uncover proteins that may play important roles in early phases of gravity signal transduction in root. Last year, we reported on the isolation and characterization of 30 proteins whose expression, modification and/or localization in root tip cells are altered within 12 min of gravity stimulation. This year, we identified 10 new differentially represented protein spots upon gravity stimulation, using a two-dimensional gel electrophoresis approach coupled with mass spectrometric identification of the corresponding proteins. This brings the total number of differentially represented protein spots to 40. We then initiated a reverse genetic analysis to better characterize the function of some of these proteins, and have been able to isolate knockout mutations in several of the corresponding genes. We also engineered Arabidopsis lines that express inverted duplications of the corresponding genes, hoping to silence both trans- and endo-genes, and generated transgenic lines that express a translational fusion between the corresponding protein and a green fluorescent polypeptide that will allow us to locate the protein within plant cells using the corresponding fluorescence. Finally, a third set of transgenic plants has been generated, which over-express several of the gravityresponding proteins. Phenotypic and cell biological analysis of these lines is underway, and should provide important clues on their role in gravity signal transduction in the root tip.<br>We also continued our analysis of one of these gravity-responding protein spots: adenosine kinase (ADK). This enzyme contributes to adenosine salvaging in cells, thereby functioning in the biochemical methyl-donor (AdoMet) cycle, polyamines and ethylene production, metabolism of multiple plant growth regulators, cell signaling and regulation of gene expression. Cell biological analyses revealed that ADK changes its biochemical properties upon gravity stimulation: Upon 12 min of stimulation, more ADK fractionates in a tris-buffer soluble fraction than before stimulation. Subsequently, ADK becomes associated with the nucleus within 30 min, possibly to salvage enough adenosine for utilization in the polyadenylation reactions that accompany dramatic gravity-induced changes in gene expression. Using reverse genetic strategies, we also demonstrated that ADK1 is needed for root gravitropism, and contributes to early phases of gravity signal transduction in the gravity-perceiving cells of the cap. There, ADK is required for proper accumulation of the plant growth regulator auxin and for its redistribution across the cap upon gravity stimulation. It is also needed for gravityinduced re-localization of the PIN3 auxin efflux facilitator to the bottom membrane of the statocytes, explaining its contribution to lateral auxin transport across the cap. On the other hand, ADK1 is not needed for gravity-induced alkalinization of the statocytes' cytoplasm, suggesting that PIN3 redistribution and cytoplasmic alkalinization are outputs of distinct pathways in those cells.<br><br>In addition to functioning in early phases of gravity signal transduction in Arabidopsis root tips, ADK also contributes to later phases of gravitropism (basipetal auxin transport, and differential cellular growth responsible for gravitropic curvature at the elongation zone). Interestingly, preliminary results suggest that specific polyamines and cytokinin contribute to these phases of the process.<br><br>Our results are important for several reasons:<br><br>1) They allowed identification of new signal transducers in Arabidopsis root tips;<br><br>2) They implicate ADK and adenosine as probable players in gravity signal transduction in root-cap statocytes. Interestingly, extra-cellular adenosine plays important signaling roles in animal systems, where it controls vesicular trafficking in neurotransmission, amongst other things. Because PIN3 re-localization involves alterations in the trafficking of vesicles within the statocytes, it will be interesting to determine if adenosine might play similar roles in plant systems;<br><br>3) Mutation in ADK1 uncouples PIN3 re-localization from cytoplasmic alkalinization in gravity-stimulated root-cap statocytes. Hence, these two physiological outputs of gravity signal transduction appear to result from distinct pathways;<br><br>4) ADK is an enzyme that uses ATP to salvage adenosine. Hence, its contribution togravity signal transduction may be sensitive to the energy status of the statocytes. An important corollary of this conclusion is that roots may be able to decrease their sensitivity to gravity stimulation under stressful conditions, thereby favoring other tropistic systems that might be more efficient at guiding them toward more favorable environments; |

| | |
|---|---|
| | 5) Some of the new signal transducers identified in our studies are also involved in other important aspects of plant growth and development;<br><br>6) Hence our research points to an intricate and yet poorly understood connection between gravity signal transduction and other essential cell biological processes in plants, calling for greater caution in the design and engineering of optimized plants for utilization in bioregenerative life support systems for long-term, manned, space exploration missions. More extensive basic research is needed, such that better informed engineering strategies can be designed to decrease the risk of catastrophic system breakdown at the midst of a mission. |
| **Bibliography Type:** | Description: (Last Updated: 07/03/2024) |
| **Abstracts for Journals and Proceedings** | Harrison, B.R.; Masson, P.H. "Investigating the Function of ARG1 and ARL2 in the Arabidopsis root gravitropic response" Abstract 29, Annual Meeting of the American Society of Gravitational and Space Biology, New York, NY, November 9-12, 2004<br>Gravit Space Biol Bull 18(1), 22 , Nov-2004 |
| **Abstracts for Journals and Proceedings** | Perrin, P.; Will, J.; Wang, Y.; Yuen, C.; Masson, P.H. "WVD2, A Novel Microtubule Binding Protein Affecting Plant Anisotropic Cell Expansion, Morphology and Helical Handedness" Abstract 178, 16th International Conference on Arabidopsis Research, Madison, WI, June 15-19, 2005<br>Book of Abstracts, University of Wisconsin-Madison , May-2005 |
| **Abstracts for Journals and Proceedings** | Young, L.-S.; Masson, P.H. "Proteomics Approach Identifies Adenosine Kinase as A Component in Arabidopsis Root Gravitropism" Abstract 59, Annual Meeting of the American Society of Gravitational and Space Biology, New York, NY, November 9-12, 2004<br>Gravit Space Biol Bull 18(1), 22 , Nov-2004 |
| **Abstracts for Journals and Proceedings** | Wang, Y.; Will, J.; Perrin, R.; Yuen, C.; Masson, P.H. "WVD2-Like Proteins Regulate Plant Growth Behavior and Anisotropic Cell Expansion " Abstract 381, 16th International Conference on Arabidopsis Research, Madison, WI, June 15-19, 2005<br>Book of Abstracts, University of Wisconsin-Madison , May-2005 |
| **Abstracts for Journals and Proceedings** | Harrison, B.R.; Masson, P.H. "Investigating the Function of ARG1 and ARL2 in the Root Gravitropic Response of Arabidopsis" Abstract 477, 16th International Conference on Arabidopsis Research, Madison, WI, June 15-19, 2005<br>Book of Abstracts, University of Wisconsin-Madison, June 15-19, 2005 , May-2005 |
| **Abstracts for Journals and Proceedings** | Murthy, N.U.M.; Young, L.S., Sabat, G.; Masson, P.H. "A Systematic Proteomic Approach to Study Early Phases of Root Gravitropism in Arabidopsis thaliana " Abstract 648, 16th International Conference on Arabidopsis Research, Madison, WI, June 15-19, 2005<br>Book of Abstracts, University of Wisconsin-Madison , May-2005 |
| **Articles in Peer-reviewed Journals** | Peer, W.; Bandyopadhyah, A.; Blakeslee, J.; Makam, S.; Chen, R.; Masson, P.H. Murphy, A. "Variation in Expression and Protein Localization of the PIN Family of Auxin Efflux Facilitator Proteins in Flavonoid Mutants with Altered Auxin Transport in Arabidopsis thaliana. " Plant Cell 16(7): 1898-1911. , Jul-2004 |
| **Articles in Peer-reviewed Journals** | Shin, H.; Shin, H.-S.; Guo, Z.; Blancaflor, E.B.; Masson, P.H.; Chen, R. "Complex Regulation of Arabidopsis AGR1/PIN2-Mediated Root Gravitropic Response and Basipetal Auxin Transport by Cantharidin-Sensitive Protein Phosphatases. " Plant J 42(2): 188-200. , Apr-2005 |
| **Articles in Peer-reviewed Journals** | Yuen, C.Y.L.; Sedbrook, J.C.; Perrin, R.M.; Carroll, K.L.; Masson, P.H. "Loss-of-Function Mutations of ROOT HAIR DEFECTIVE3 Suppress Root Waving, Skewing, and Epidermal Cell File Rotation in Arabidopsis. " Plant Physiol. 138(2):701-714 , Jun-2005 |
| **Articles in Peer-reviewed Journals** | Perrin, R.M.; Young, L-S.; Murthy UM, N.; Harrison, B.R.; Wang, Y.; Will, J.L.; Masson, P.H. "Gravity Signal Transduction in Primary Roots. " Ann Bot 96: 737-743 , Sep-2005 |
| **Books/Book Chapters** | Chen, R.; Masson, P.H. "Auxin Transport and Recycling of PIN Proteins in Plants" in "Plant Cell Monog" Ed. Baluska, F. Springer-Verlag, 1-19 , Nov-2005 |
| **Presentation** | Masson, P.H. "Gravity Signal Tranduction in Arabidopsis Roots" Plant Biology Seminar, North Carolina State University<br>Mar-2005 |
| **Presentation** | Masson, P.H.; Young, L.-S.; Murthy, N.U.M.; Sabat, G.; Harrison, B.R.; Stanga, J.; Neal, C.; Vaughn, L. "Genetic and Proteomic Analysis of Gravity Signal transduction in Arabidopsis roots" Keynote Speaker, XVIIth International Botanical Congress, Austria Center Vienna, Austria<br>Jul-2005 |
| **Presentation** | Young, L.-S.; Masson, P.H. "A Mutation in The Arabidopsis ADK1 Gene Affects Root Gravitropism, Columella Morphogenesis and Lateral Auxin Transport Across the Root Tip" 16th Conference on Arabidopsis Research, Madison, WI, June 16, 2005<br>Jun-2005 |
| **Presentation** | Masson, P.H.; Young, L.-S.; Murthy, N.U.M.; Sabat, G.; Harrison, B.R.; Stanga, J.; Neal, C.; Vaughn, L. "Genetic and Proteomic Analysis of Gravity Signal transduction in Arabidopsis roots" Gordon Research Conference on Mechanotransduction and Gravity Signaling in Biological Systems, July 26, 2005<br>Jul-2005 |
| **Presentation** | Masson, P.H. "Genetic and Proteomic Analysis of Gravity Signal transduction in Arabidopsis roots" 2005/6 Syracuse University Biology Lecture Series<br>Oct-2005 |